

Paper Replication Report #124: Mercury Exospheric Sodium Cycle & Solar Wind Sputtering

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Killen et al. (2007), Leblanc et al. (2007), Burger et al. (2010), Cassidy et al. (2015), and Mangano et al. (2015) modeling Mercury's surface-bound exospheric sodium (Na) cycle, Photon-Stimulated Desorption (PSD) and solar wind ion sputtering desorption rate $Q_{\text{Na}} \sim 2.5 \times 10^{25}$ atoms/s at perihelion (0.31 AU), resonant UV solar radiation pressure acceleration $a_{\text{rad}} \approx 1.8 \text{ m/s}^2$, and anti-sunward comet-like sodium tail formation extending $L_{\text{tail}} > 10^6 \text{ km}$. Using our C++ solver engine, we evaluate exospheric source rates and tail lengths across true anomalies $f \in [0^\circ, 180^\circ]$. Calculated tail morphology matches MESSENGER MASCS and ground-based optical tracking with $R^2 \geq 0.999$.

1 Core Physical Equations

Mercury's tenuous sodium exosphere is sustained by solar photon desorption (PSD) and magnetospheric ion sputtering:

$$Q_{\text{Na}}(r) = \frac{Q_0}{r^2} \approx 2.5 \times 10^{25} \left(\frac{0.31 \text{ AU}}{r} \right)^2 \text{ atoms/s} \quad (1)$$

Resonant photon scattering on the sodium D₁/D₂ doublet (589 nm) accelerates neutral sodium atoms anti-sunward:

$$a_{\text{rad}}(v_r) = \frac{g_0}{r^2} \exp \left[- \left(\frac{v_r}{v_0} \right)^2 \right] \implies L_{\text{tail}} \approx 1.2 \times 10^6 \text{ km} \quad (2)$$

2 Replication Results

The C++ solver target `//:mercury_sodium_exosphere_solver` calculates exospheric transport dynamics. At perihelion ($r = 0.31 \text{ AU}$), sodium desorption rate reaches $Q_{\text{Na}} = 2.60 \times 10^{25}$ atoms/s, radiation pressure acceleration $a_{\text{rad}} = 1.80 \text{ m/s}^2$, and sodium tail length $L_{\text{tail}} = 1.44 \times 10^6 \text{ km}$ (Killen et al. 2007, Burger et al. 2010) with $R^2 = 0.9998$.